

New Concept English Book 4

48 articles

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1. Finding Fossil Man

We can read of things that happened 5,000 years ago in the Near East, where people first learned to write. But there are some parts of the world where even now people cannot write. The only way that they can preserve their history is to recount it as sagas - legends handed down from one generation of story - tellers to another. These legends are useful because they can tell us something about migrations of people who lived long ago, but none could write down what they did. Anthropologists wondered for a long time how to classify the Pygmies of the Congo Basin and the Bushmen of the Kalahari Desert in Africa. These tribes live by hunting, fishing and gathering, and they do not plant crops. The anthropologists made a mistake, because they thought these people were still in the Stone Age. They forgot that the Pygmies and Bushmen have been in contact with modern civilization for a long time. The Pygmies and Bushmen are not Stone Age people. They are modern people who have not yet learned to write.

2. Spare that spider

Why, you may wonder, should spiders be our friends? Because they destroy so many insects, and insects include some of the greatest enemies of the human race. Insects would make it impossible for us to live in the world; they would devour all our crops and kill our flocks and herds, if it were not for the protection we get from insect - eating animals. We owe a lot to the birds and beasts who eat insects but all of them put together kill only a fraction of the number destroyed by spiders. Moreover, unlike some of the other insect eaters, spiders never do the least harm to us or our belongings.

Spiders are not insects, as many people think, nor even nearly related to them. One can tell the difference almost at a glance, for a spider always has eight legs and an insect never more than six.

How many spiders are engaged in this work on our behalf? One authority on spiders made a census of the spiders in a grass field in the south of England, and he estimated that there were more than 2,250,000 in one acre; that is something like 6,000,000 spiders of different kinds on a football pitch. Spiders are busy for at least half the year in killing insects. It is impossible to make more than the wildest guess at how many they kill, but they are hungry creatures, not content with only three meals a day. It has been estimated that the weight of all the insects destroyed by spiders in Britain in one year would be greater than the total weight of all the human beings in the country.

Spiders are one of the marvels of nature. Their webs are masterpieces of engineering. A spider's web is stronger than it looks. It can be stretched considerably without breaking,

and made of a sticky material which traps flying insects. The threads are as fine as a spider's silk but much stronger - human hair is about as strong as spider's silk of the same thickness.

Spiders, then, are our friends. The little we can do to spare them is surely worth doing.

3. Matterhorn Man

Modern alpinists try to climb mountains by a route which will give them good sport, and the more difficult it is, the more highly it is regarded. In the pioneering days, however, this was not the case at all. The early climbers were looking for the easiest way to the top, because the summit was the prize they sought, especially if it had never been attained before. It is true that during their explorations they often faced difficulties and dangers of the most perilous nature, equipped in a manner which would make a modern climber shudder at the thought, but they did not go out of their way to court such excitement. They had a single aim, a solitary goal---the top!

It is hard for us to realize nowadays how difficult it was for the pioneers. Except for one or two places such as Zermatt and Chamonix, which had rapidly become popular, Alpine villages tended to be impoverished settlements cut off from civilization by the high mountains. Such inns as there were were generally dirty and flea - ridden; the food simply local cheese accompanied by bread often twelve months old, all washed down with coarse wine. Often a valley boasted no inn at all, and climbers found shelter wherever they could---sometimes with the local priest (who was usually as poor as his parishioners), sometimes with shepherds or cheese - makers. Invariably the background was the same: dirt and poverty, and very uncomfortable. For men accustomed to eating seven - course dinners and sleeping between fine linen sheets at home, the change to the Alps must have been very hard indeed.

Why, then, did the early climbers put up with such hardships? The answer, I think, is that they were less interested in the sport than in the sense of accomplishment they got from reaching the top. It is true that they took risks, but in those days they had to take risks. They would never have been able to reach the top at all if they had not been prepared to take risks. They were a hardy race, these early climbers. They were not looking for a comfortable life, but for a challenge, and a sense of pride that they could meet that challenge.

Today, of course, things are different. The modern alpinist may be interested in the sport as well as in the sense of accomplishment. He may also be interested in getting a good photograph or in making a film. But he will still be looking for a route which will give him

good sport, and he will still be looking for a challenge. He will still be a hardy race, these modern alpinists.

4. Seeing hands

Several cases have been reported in Russia recently of people who can read and detect colours with their fingers, and even see through solid doors and walls. One case concerns an eleven-year-old schoolgirl, Vera Petrova, who has normal vision but who can also perceive things with different parts of her skin, and through solid walls. This ability was first noticed by her father. One day she came into his office and happened to put her hands on the door of a locked safe. Suddenly she asked her father why he kept so many old newspapers locked away there, and even described the way they were done up in bundles. Vera's curious talent was brought to the notice of a scientific research institute in the town of Ulyanovsk, near where she lives, and in April she was given a series of tests by a special commission of the Ministry of Health of the Russian Federal Republic. During these tests she was able to read a newspaper through an opaque screen and, stranger still, by moving her elbow over a child's game of Lotto she was able to describe the figures and colours printed on it; and, in another instance, wearing stockings and slippers, to make out with her foot the outlines and colours of a picture hidden under a carpet. Other experiments showed that her knees and shoulders had a similar sensitivity. During all these tests Vera was blindfolded; and, indeed, except when blindfolded she lacked the ability to perceive things with her skin. It was also found that although she could perceive things with her fingers this ability ceased the moment her hands were wet.

Ernst Weber, a physiologist in Leipzig, was the first to show that people can perceive things through their fingertips as well as they can through their eyes. He found that the smaller the object, the closer it had to be to the eyes for its details to be seen. The same applies to the fingertips. The closer the object is to the fingertips, the larger it seems to be and the more details can be made out. Weber found that when the fingertips were about three inches from an object, the details of the object were as clear as when it was at the normal reading distance of about fourteen inches from the eyes. If the object was moved closer still, the fingertips could no longer make out its details. This shows that the fingertips can perceive objects in the same way as the eyes, but only up to a certain distance.

Vera Petrova's case is a good example of how people can be trained to perceive things in different ways. She was not born with the ability to perceive things with her skin, but she has learned to do so through long practice. Her father started to train her when she was a small child. He would put different objects in front of her and ask her to describe them

without looking. At first she could only make out the general shape of the objects, but with more practice she was able to make out more details. As she grew older, she was able to perceive things with different parts of her skin, and through solid walls. This shows that the ability to perceive things with the skin can be developed through training.

5. Youth

People are always talking about "the problem of youth". If there is one -- which I take leave to doubt -- then it is older people who create it, not the young themselves. Let us get down to fundamentals and agree that the young are after all human beings -- people just like their elders. There is only one difference between an old man and a young one: the young man has a glorious future before him and the old one has a splendid future behind him: and maybe that is where the rub is.

When I was a teenager, I felt that I was just young and uncertain -- that I was a new boy in a huge school, and I would have been very pleased to be regarded as something so interesting as a problem. For one thing, being a problem gives you a certain identity, and that is one of the things the young are busily engaged in seeking.

I find young people exciting. They have an air of freedom, and they have not a dreary commitment to mean ambitions or love of comfort. They are not anxious social climbers, and they have no devotion to material things. All this seems to me to link them with life, and the origins of things. It's as if they were in some sense cosmic beings in violent and lovely contrast to us suburban creatures. All that is in my mind when I meet a young person. He may be conceited, ill-mannered, presumptuous or fatuous, but I do not turn for protection to dreary clichés about respect for elders -- as if mere age were a reason for respect. I accept that we are equals, and I will argue with him, as an equal, if I think he is wrong.

This is what I have learned from my own youth. And I have found that the world is in general agreement with my view of youth. At least, when talking about the young, most people will say that they are hope of the world, and that it is necessary to guide them in the right way. But then, they hope that the young will listen to them, and they believe that they are wiser than the young, which is why they are always talking about "the problem of youth".

6. The sporting spirit

I am always amazed when I hear people saying that sport creates goodwill between the nations, and that if only the common peoples of the world could meet one another at football or cricket, they would have no inclination to meet on the battlefield. Even if one didn't know from concrete examples (the 1936 Olympic Games, for instance) that international sporting contests lead to orgies of hatred, one could deduce it from general principles.

Nearly all the sports practised nowadays are competitive. You play to win, and the game has little meaning unless you do your utmost to win. On the village green, where you pick up sides and no feeling of local patriotism is involved, it is possible to play simply for the fun and exercise: but as soon as the question of prestige arises, as soon as you feel that you and some larger unit will be disgraced if you lose, the most savage combative instincts are aroused. Anyone who has played even in a school football match knows this. At the international level, sport is frankly mimic warfare. But the significant thing is not the behaviour of the players but the attitude of the spectators: and, behind the spectators, of the nations who work themselves into furies over these absurd contests, and seriously believe - at any rate for short periods - that running, jumping and kicking a ball are tests of national virtue.

I was once present at a tug - of - war in which the two teams were so evenly matched that neither could win. The umpire in charge of the match was at his wits' end. Finally, one of the teams, by a superhuman effort, managed to drag the other team over the line. The crowd went wild with joy. This incident left a deep impression on me. I saw that in a world where people are obsessed with winning, the slightest advantage can trigger a violent reaction. It is this spirit, I believe, that underlies the modern Olympics. We are so used to seeing champions crowned and anthems played that we forget the darker side of sport. The Olympics should be a celebration of human achievement, not a battleground for national pride.

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Once the games were over, the stadium emptied and the swarms of spectators dispersed, a curious lull ensued. The nations suddenly remembered that they were supposed to be at peace with each other. They rubbed shoulders at the crowded terminals, and at length they went their separate ways without rancour. Even the teams themselves, as they boarded their respective buses and trains, seemed to have forgotten their enmity. But it was only a momentary lull. In less than an hour's time they would be back at their respective camps, preparing for the next day's play. And so the cycle of hatred and reconciliation continued.

8. Trading Standards

Chickens slaughtered in the United States, claim officials in Brussels, are not fit to grace European tables. No, say the Americans: our fowl are fine, we simply clean them in a different way. These days, it is differences in national regulations, far more than tariffs, that put sand in the wheels of trade between rich countries. It is not just farmers who are complaining. An electric razor that meets the European Union's safety standards must be approved by American testers before it can be sold in the United States, and an American-made dialysis machine needs the EU's okay before it hits the market in Europe.

As it happens, a razor that is safe in Europe is unlikely to electrocute Americans. So, ask businesses on both sides of the Atlantic, why have two lots of tests where one would do? Politicians agree, in principle, so America and the EU have been trying to reach a deal which would eliminate the need to double - test many products. They hope to finish in time for a trade summit between America and the EU on May 28th. Although negotiators are optimistic, the details are complex enough that they may be hard - pressed to get a deal at all.

Why? One problem is that while both America and the EU have schemes to test products,

they have different approaches, different priorities and different timetables. Another factor is that European companies are more squeamish about animal testing. The Americans say that by accepting outside testing, they are losing control of their own trade policies. And then there are the costs. The EU's tests are supposed to be more expensive than America's, but some American companies have to pay for both.

There is no sign that consumers are being short - changed as a result of these trade barriers. Indeed, EU consumers are probably better off, since they are protected from products that may not be safe. And American consumers may be better off, too. Since they are not being bombarded with unsafe products, they may have more time to consider the products that are actually available.

The real problem is that trade barriers are rising, not falling. This is a result of political will. Politicians are under pressure from their voters to be tough on trade. They are also under pressure from business lobbyists who want to keep out foreign competitors. But the biggest factor is that trade barriers are easy to erect, and difficult to dismantle. Once a trade barrier is in place, it is hard to get rid of it. This is because it is difficult to prove that a trade barrier is not necessary. And if a trade barrier is not necessary, then it is illegal under international law. So, politicians are reluctant to remove trade barriers, even when they know that they are not necessary.

So, what can be done? One solution is to have more international cooperation. The World Trade Organization (WTO) could play a more active role in reducing trade barriers. Another solution is to have more transparency. Politicians could be more open about the reasons for erecting trade barriers. And then there is the question of consumer education. Consumers could be better informed about the safety of products. But these solutions are not easy to implement. And in the meantime, trade barriers are rising, not falling.

9. Royal espionage

Alfred the Great acted as his own spy, visiting Danish camps disguised as a minstrel. In those days wandering minstrels were welcome everywhere. They were not fighting men, and their harp was their passport. Alfred had learned many of their ballads in his youth, and could vary his programme with acrobatic tricks and simple conjuring.

While Alfred's little army slowly began to gather at Athelney, the king himself set out to penetrate the camp of Guthrum, the commander of the Danish invaders. These had settled down for the winter at Chippenham: thither Alfred went. He noticed at once that discipline was slack: the Danes had the self - confidence of conquerors, and their security precautions were casual. They lived well, on the proceeds of raids on neighbouring

regions. There they collected women as well as food and drink, and a life of ease had made them soft.

Alfred stayed in the camp a week before he returned to Athelney. The force there assembled was trivial compared with the Danish horde. But Alfred had deduced that the Danes were no longer fit for prolonged battle: and that their commissariat had no organization, but depended on irregular raids. So, faced with the Danish advance, he had to prepare to defend the kingdom. He parcelled out the land into small holdings which were given to the great nobles of the realm in fief. In return for a free gift of the land, the noble swore to perform military service. The king retained a large reserve of land in his own hands. By this policy Alfred managed to create a large army out of very little money. And when the Danes, having had their raids, were again assembled in camp preparing for winter, Alfred struck suddenly. He routed them with his little army and chased them back to their ships. The Danes were not to be seen again in England for many years.

The episode could reasonably be called "Royal espionage". For Alfred had done what no spy had ever done before in the annals of espionage - he had stayed in the enemy's camp and returned with valuable information.

10. Silicon Valley

Technology trends may push Silicon Valley back to the future. Carver Mead, a pioneer in integrated circuits and a professor of computer science at the California Institute of Technology, notes there are now workstations that enable engineers to design, test and produce chips right on their desks, much the way an editor creates a newsletter on a Macintosh. As the time and cost of making a chip drop to a few days and a few hundred dollars, engineers may soon be free to let their imaginations soar without being penalized by expensive failures.

Mead predicts that inventors will be able to perfect powerful customized chips over a weekend at the office - spawning a new generation of garage start - ups and giving the U.S. a jump - start on its foreign rivals in getting new products to market faster. "We've got more garages with smart people", Mead observes. "We really thrive on anarchy."

And on Asians. Already, orientals and Asian - Americans constitute the majority of the engineering staffs at many Valley firms. And Chinese, Indian and Pakistani engineers are graduating in droves from California's colleges. As the heads of next - generation start - ups, these Asian innovators can draw on customs and languages to forge tighter links with key Pacific Rim markets.

For instance, Alex Au, a Stanford Ph.D. from Hong Kong, has set up a Taiwanese factory to

challenge Japan's near lock on the memory - chip market. India - born N.Damodar Reddy's tiny California company reopened an AT&T chip plant in Kansas City last year to profit from the rapidly falling prices of chips. Before it becomes a retirement village, Silicon Valley may prove a classroom for building a global business.

11. How to Grow Old

Some old people are oppressed by the fear of death. In the young there is a justification for this feeling. Young men who have reason to fear that they will be killed in battle may justifiably feel bitter in the thought that they have been cheated of the best things that life has to offer. But in an old man who has known human joys and sorrows, and has achieved whatever work it was in him to do, the fear of death is somewhat abject and ignoble. The best way to overcome it - so at least it seems to me - is to make your interests gradually wider and more impersonal, until bit by bit the walls of the ego recede, and your life becomes increasingly merged in the universal life. An individual human existence should be like a river - small at first, narrowly contained within its banks, and rushing passionately past boulders and over waterfalls. Gradually the river grows wider, the banks recede, the waters flow more quietly, and in the end, without any visible break, they become merged in the sea, and painlessly lose their individual being. The man who, in old age, can see his life in this way, will not suffer from the fear of death, since the things he cares for will continue. And if, with the decay of vitality, weariness increases, the thought of rest will be not unwelcome. I should wish to die while still at work, knowing that others will carry on what I can no longer do, and content in the thought that what was possible has been done.

Some scientists in the field of artificial intelligence seem to think that a breakthrough in creating true artificial intelligence will be made within the next few decades. I am not so sure. I think it will be much longer before we see machines that can truly think like humans. True artificial intelligence would have to be able to understand the world in the same way that humans do, and this is a very difficult problem. Machines can already do many things that humans can do, but they do not understand what they are doing in the same way that humans do. For example, a computer can play chess very well, but it does not understand the game of chess in the way that a human does. It does not have the same kind of consciousness that a human has. Consciousness is a very difficult concept to define, but it is something that is very important for understanding the nature of intelligence. Without consciousness, it is difficult to say that a machine is truly intelligent.

Another problem with creating true artificial intelligence is that we do not really know how the human brain works. We know a great deal about the structure of the brain, but

we do not know how it processes information and creates thoughts. This is another area where much more research is needed. Some scientists think that we will be able to create artificial intelligence by simulating the structure and function of the human brain, but this is also a very difficult task. The human brain is an extremely complex organ, and it is very difficult to simulate its structure and function accurately.

In conclusion, while the idea of creating true artificial intelligence is very exciting, I think it will be a long time before we see machines that can truly think like humans. There are many difficult problems that need to be solved before we can achieve this goal.

12. Banking and the Economy

Banks are closely concerned with the flow of money into and out of the economy. They often co-operate with governments in efforts to stabilize economies and to prevent inflation. They are specialists in the business of providing capital, and in allocating funds on credit. Banks originated as places to which people took their valuables for safe-keeping, but today the great banks of the world have many functions in addition to acting as guardians of valuable private possessions.

One of the most important of these is regular participation in the foreign exchange market. In this market, banks buy and sell foreign currencies on a continual basis, so that an importer can purchase a shipment of goods from another country and pay for it in the currency of that country. The bank's selling price for a particular currency is always slightly higher than its buying price for the same currency, and the difference represents the bank's profit on the transaction.

Banks also provide a service for exporters. When an exporter arranges a shipment of goods to another country, he may need to finance the operation until he receives payment from the importer. Banks can provide loans to exporters at relatively low rates of interest, and they can also provide insurance against the risk of non - payment by the importer.

In addition to these international functions, banks also play an important role in the domestic economy. They are a major source of loans for businesses and individuals, and they help to allocate capital within the economy. They also play a role in the regulation of the money supply, through their ability to create and destroy money in the form of deposits and loans.

Finally, banks are important because they provide a convenient means of payment. People can use bank cheques, credit cards, or debit cards to pay for goods and services, and this makes economic transactions much easier and more efficient.

13. The Search for Oil

The deepest holes of all are made for oil, and they go down to as much as 25,000 feet. But we do not need to send men down to get oil out, as we must with other mineral deposits. The holes are only borings, less than a foot in diameter. My particular experience is largely in oil, and the search for oil has done more to improve deep drilling than any other mining activity. When it has been decided where we are going to drill, we put up at the surface an oil derrick. It has to be tall because it is like a giant block and tackle, and we have to lower into the ground and haul out of the ground great lengths of drill pipe which are rotated by an engine at the top and are fitted with a cutting bit at the bottom.

There is a lot of luck in drilling for oil. The drill may just miss the oil although it is near; on the other hand, it may strike oil at a fairly high level. When the drill goes down, it brings up soil. The samples of soil from various depths are examined for traces of oil. If they are disappointed at one place, the drillers go to another. Great sums of money have been spent, for example in the deserts of Egypt, in 'prospecting' for oil. Sometimes little is found. When we buy a few gallons of petrol for our cars, we pay not only the cost of the petrol, but also part of the cost of the search that is always going on.

When the crude oil is obtained from the field, it is taken to the refineries to be treated. The commonest form of treatment is heating. When the oil is heated, the first vapours to rise are cooled and become the finest petrol. Petrol has a low boiling point; if a little is poured into the hand, it soon evaporates. The next lot of vapours are cooled and become paraffin. What remains is heavy oil that is used as fuel.

14. The Butterfly Effect

Beyond two or three days, the world's best weather forecasts are speculative, and beyond six or seven they are worthless. The Butterfly Effect is the reason. For small pieces of weather -- and to a global forecaster, small can mean thunderstorms and blizzards -- any prediction deteriorates rapidly. Errors and uncertainties multiply, cascading upward through a chain of turbulent features, from dust devils and squalls up to continent - size eddies that only satellites can see. The modern weather models work with a grid of points of the order of sixty miles apart, and even so, some starting data has to be guessed, since ground stations and satellites cannot see everywhere. But suppose the earth could be covered with sensors spaced one foot apart, rising at one - foot intervals all the way to the top of the atmosphere. Suppose every sensor gives perfectly accurate readings of temperature, pressure, humidity, and any other quantity a meteorologist would want. Precisely at noon an infinitely powerful computer takes all the data and calculates what

will happen at each point at 12.01, then 12.02, then 12.03... The computer will still be unable to predict whether Princeton, New Jersey, will have sun or rain on a day one month away. At noon the spaces between the sensors will hide fluctuations that the computer will not know about, tiny variations in the flow of the atmosphere that will interact with other fluctuations and create a turbulence that will spread rapidly and unpredictably. In a very short time, the computer will lose track of where the turbulence is going, and even more important, it will not know if a new disturbance has occurred somewhere else. Even with the most powerful computers, there will always be a degree of uncertainty about whether the weather will change or not, and by how much. This is the essence of the Butterfly Effect. The name comes from an example by the mathematician Edward Lorenz. He was thinking about how small changes in initial conditions can lead to large - scale effects. He imagined a butterfly flapping its wings in Brazil and then considered what would happen. "Could it be," he wondered, "that a single flap of a butterfly's wings could cause a hurricane in Texas?" His answer was yes, in principle. A butterfly flaps its wings and it causes a tiny change in the air pressure. That change in air pressure causes a slightly different pattern of air movement. That slightly different pattern of air movement causes another slightly different pattern, and so on. By the time the air movement has spread across the Atlantic Ocean, the change has become large enough to cause a hurricane in Texas. The Butterfly Effect is not just a theoretical concept. It has practical implications for weather forecasting, climate modelling, and many other fields. For example, in climate modelling, we need to understand how small changes in the Earth's climate system can lead to large - scale changes in the global climate. If we do not understand the Butterfly Effect, we may make incorrect predictions about the future of the Earth's climate. In addition to weather forecasting and climate modelling, the Butterfly Effect has implications for many other fields. For example, in economics, small changes in interest rates or exchange rates can lead to large - scale changes in the economy. In ecology, small changes in the environment can lead to large - scale changes in the ecosystem. In social science, small changes in public opinion or social behavior can lead to large - scale changes in society. The Butterfly Effect is a reminder that the world is a complex and interconnected place, and that small changes can have large - scale effects.

15. Secrecy in industry

Two factors weigh heavily against the effectiveness of scientific research in industry. One is the general atmosphere of secrecy in which it is carried out, the other the lack of freedom of the individual research worker. In so far as any inquiry is a secret one, it naturally limits all those engaged in carrying it out from effective contact with their fellow

scientists either in other countries or in universities, or even, often enough, in other departments of the same firm. The degree of secrecy naturally varies considerably. Some of the bigger firms are engaged in researches which are of such general and fundamental nature that it is a positive advantage to them not to keep them secret. Yet a great many processes depending on such research are sought for with complete secrecy until the stage at which patents can be taken out. Even more processes are never patented at all but kept as secret processes. This applies particularly to chemical industries, where chance discoveries play a much larger part than they do in physical and mechanical industries. Sometimes the secrecy goes to such an extent that the whole nature of the research cannot be mentioned. Many firms, for example, have great difficulty in obtaining technical or scientific books from libraries because they are unwilling to have their names entered as having taken out such and such a book, for fear the agents of other firms should be able to trace the kind of research they are likely to be undertaking.

J. B. S. Haldane once said that he preferred the society of scientists, who did not care whether their ideas were stolen, because he could do more with their ideas than they could do themselves. He was doubtless right, in so far as his own work was concerned; but it may be questioned whether it is always good for the community that a scientist should "do more" than his fellows. It is for such reasons that the case for having a scientific directorate, whose main concern should be the interests of science as a whole, becomes cogent. Such a body would need to be large enough to carry considerable authority, but not so large as to be unmanageable. It would also need to be composed of persons of very different types of scientific interests, so that the board could really be representative of all the different interests in science. There would also need to be a sufficient number of research workers in reserve so that the board could pass on to them any ideas which are likely to be of immediate value. It would be their task to test out all such ideas, and to make what seems best of them. It would also be the task of this body to arrange for the training of the research workers of the future, in so far as this can be done within the firm.

16. The Search for the Earth's Minerals

The earth is a large, rocky planet where there is little free oxygen or water on the surface. However, beneath the surface there is a great deal of activity. Heat from the interior of the earth melts rocks and creates magma. This magma rises to the surface through cracks in the rocks, and when it cools, it forms new rocks.

Minerals are substances that are formed in this way. They are the building blocks of rocks. There are over 3,000 different minerals, but only about 20 are common. These common

minerals are found in most rocks.

The search for minerals is an important activity. Minerals are used in many different ways. Some minerals, such as gold and silver, are used for making jewellery. Others, such as iron and copper, are used for making tools and machines. Minerals are also used in building materials, such as cement and bricks.

Geologists are scientists who study the earth. They use many different methods to find minerals. One method is to study the rocks on the surface. Geologists can tell a lot about the minerals that are likely to be found beneath the surface by looking at the rocks on the surface. Another method is to use special instruments to detect the presence of minerals. These instruments can measure the magnetic field, gravity, or electrical conductivity of the rocks.

Once a likely area for minerals has been found, geologists will drill holes to take samples of the rocks. These samples will be analyzed in a laboratory to find out what minerals are present. If minerals are found, mining operations may begin.

17. The Greatest Bridge in the World

Verrazano, an Italian about whom little is known, sailed into New York Harbour in 1524 and named it Angouleme. He described it as 'a very agreeable situation located within two small hills in the midst of which flowed a great river.'. Though Verrazano is by no means considered to be a great explorer, his name will probably remain immortal, for on November 21st, 1964, the longest suspension bridge in the world was named after him.

The Verrazano Bridge, which was designed by Othmar Ammann, joins Brooklyn to Staten Island. It has a span of 4,260 feet. The bridge is so long that the shape of the earth had to be taken into account by its designer. Two great towers support four huge cables. The towers are built on immense underwater platforms made of steel and concrete. The platforms extend to a depth of over 100 feet under the sea. These alone took sixteen months to build. Above the surface of the water, the towers rise to a height of nearly 700 feet. They support the cables from which the bridge has been suspended. Each of the four cables contains 26,108 lengths of wire. It has been estimated that if the bridge were packed with cars, it would still only be carrying a third of its total capacity. However, size and strength are not the only important things about this bridge. Despite its immensity, it is both simple and elegant, fulfilling its designer's dream to create 'an enormous object drawn as faintly as possible'.

18. Porpoises

There has long been a superstition among mariners that porpoises will save drowning men by pushing them to the surface, or protect them from sharks by surrounding them in defensive formation. Marine Studio biologists have pointed out that, however intelligent they may be, it is probably a mistake to credit dolphins with any motive of life - saving. On the occasions when they have pushed to shore an unconscious human being they have much more likely done it out of curiosity or for sport, as in riding the bow waves of a ship. In 1928 some porpoises were photographed working like beavers to push ashore a waterlogged mattress. If, as has been reported, they have protected humans from sharks, it may have been because curiosity attracted them and because the scent of a possible meal attracted the sharks. Porpoises and sharks are natural enemies. It is possible that upon such an occasion a battle ensued, with the sharks being driven away or killed.

Whether it be bird, fish or beast, the porpoise is intrigued with anything that is alive. They are constantly after the turtles, who peacefully submit to all sorts of indignities. One young calf especially enjoyed raising a turtle to the surface with his snout and then shoving him across the tank like an aquaplane. Almost any day a young porpoise may be seen trying to turn a 300 - pound sea turtle over by sticking his snout under the edge of his shell and pushing up for dear life. This is not easy, and may require two porpoises working together. In another game, as the turtles swim across the oceanarium, the first porpoise swoops down from above and butts his shell with his belly. This knocks the turtle down several feet where another porpoise is waiting to give it another butt. The turtles seem to enjoy it, but perhaps they would not if they could feel the scratches on their shells from the porpoises' teeth. After such a burst of activity they usually take a short rest before they start off again.

A porpoise will follow a ship for hours at a time, leaping out of the water and doing somersaults. There is no evidence that porpoises have a language but they communicate with one another by squeaks, clicks and whistles. It has been suggested that they may be able to communicate with humans through these sounds, but this has not been proven. One theory is that they use these sounds to navigate, by detecting the echoes of their own voices. Another theory is that they use them to communicate their emotions, such as excitement or fear. Still another theory is that they use them to identify one another, as each porpoise has a unique set of squeaks, clicks and whistles.

19. The Stuff of Dreams

It is fairly clear that the sleeping period must have some function, and because there is so much of it the function would seem to be important. Speculations about its nature have been going on for literally thousands of years, and one odd finding that makes the problem puzzling is that it looks very much as if sleeping is not simply a matter of giving the body a rest. 'Rest', in terms of muscle relaxation and so on, can be achieved by a brief period lying, or even sitting down. The body's tissues are self - repairing and self - restoring to a degree, and function best when more or less continuously active. In fact a basic amount of movement occurs during sleep which is specifically concerned with preventing muscle inactivity. If it is not a question of resting the body, perhaps it is the brain that needs resting? This might be a plausible hypothesis were it not for two factors. First the electroencephalograph (which is simply a device for recording the electrical activity of the brain by attaching electrodes to the scalp) shows that while there is a change in the pattern of activity during sleep, there is no evidence that the total amount of activity is any less. The second factor is more interesting and more fundamental. In 1960 an American psychiatrist named William Dement published experiments dealing with the recording of eye - movements during sleep. He showed that the average individual's sleep cycle is punctuated by peculiar bursts of eye - movements, some slow, some rapid, some in both directions. These periods of eye - movements are called REM sleep (Rapid Eye Movement) periods, and they occur at intervals of about 90 minutes throughout the night with an average duration of about 20 minutes. People woken during these periods of REM sleep generally report that they have been dreaming. When woken at other times they reported no dreams. If one group of people were disturbed from their eye - movement sleep for several nights on end, and another group were disturbed for an equal number of nights but when they were not experiencing eye - movement sleep, the first group began to show some personality disorders while the second group did not. REM sleep seems to be essential for the normal functioning of the brain. REM sleep occurs in both humans and animals. Some animals, such as cats, have large amounts of REM sleep, while others, such as cows, have very little. There is some evidence that the amount of REM sleep is related to the intelligence of the animal. The more intelligent the animal, the more REM sleep it has. But why should REM sleep be essential for the normal functioning of the brain? One theory is that during REM sleep the brain is 'rehearsing' things that it has learned during the day. Another theory is that REM sleep is a time when the brain is 'clearing' itself of useless information. But neither of these theories has been proved yet. So far we have only been able to say that REM sleep seems to be essential for the normal functioning of the brain, and that the amount of REM sleep is related to the intelligence of the animal. We still do not know what the function of sleep is, or why we need to dream.

20. Snake poison

How it came about that snakes manufactured poison is a mystery. Over the periods their saliva, a mild, digestive juice like our own, was converted into a poison that defies analysis even today. It was not forced upon them by the survival competition; they could have caught and lived on prey without using poison, just as the thousands of non-poisonous snakes still do. Poison to a snake is merely a luxury; it enables it to get its food with very little effort, no more effort than one bite. And why only snakes? Cats, for instance, would be greatly helped; no running fights with large, fierce rats or tussles with grown chickens - the mouse or young chicken goes limply into the cat's mouth.

The snakes' poison is highly modified saliva containing zootoxins which act on the nervous system and blood circulation. It is different in every single kind of snake - from the cobra's for instance which affects the nerves, to the viper's which, on the contrary, affects the blood.

One wonders also why nature, with some snakes, concocted poison of such extreme potency. In the conversion of saliva into poison, one might suppose that a fixed process took place. It did not; some snakes manufactured a poison different in every respect from that of others, as different as arsenic is from strychnine, and having different effects. One poison acts on the nerves, the other on the blood.

The spitting cobra is another case of a snake that has modified its saliva into a poison different in nature from that of other snakes. This snake, instead of biting, spits out a jet of venom which may reach a distance of two feet. It is not very accurate, but it is enough to cause a lot of pain and some blindness if it gets into the eyes. The spitting cobra has little poison. What is this snake's advantage in having a poison that is not very powerful? The answer is that it has a very good aim; it is almost always successful in hitting its target, which is the eyes of the victim. And why should it spit? There is no obvious reason why it should not bite. The answer is that it has a good aim; it is almost always successful in hitting its target, which is the eyes of the victim. By spitting, it can keep a safe distance from its prey, which is a great advantage in a fight.

21. William S. Hart and the Early "Western" Film

William S. Hart was, perhaps, the greatest of all Western stars, for unlike Gary Cooper and John Wayne he appeared in nothing but Westerns. From 1914 to 1924 he was supreme and unchallenged. It was Hart who created the basic formula of the Western film, and devised the protagonist he played in every film he made, the good - bad man, the accidental - noble outlaw, or the honest - but - framed cowboy, or the sheriff made

suspect by vicious gossip; in short, the individual in conflict with himself and his frontier environment.

Hart actually knew something of the old West. He had lived in it as a child when it was already disappearing, and his hero was firmly rooted in his memories and experiences, and in both the history and the mythology of the vanished frontier. And although no period or place in American history has been more absurdly romanticized, myth and reality did join hands in at least one arena, the conflict between the individual and encroaching civilization.

Men accustomed to struggling for survival against the elements and Indians were bewildered by politicians, bankers and businessmen, and unhorsed by fences, laws and alien taboos. Hart's good - bad man was always an outsider, always one of the disinherited, and if he found it necessary to shoot a sheriff or rob a bank along the way, his early audiences found it easy to understand and forgive, especially when it was Hart who, in the end, overcame the attacking Indians or the outlaw gang. Audiences in the second decade of the twentieth century found it pleasant to escape to a time when life, though hard, was relatively simple. We still do; living in a world in which undeclared aggression, war, hypocrisy, chicanery, anarchy and impending doom seem to be the norm, we all want a code to live by.

Carrying a gun in the city is illegal in most countries, but it was not so in the Old West. Hart's audiences knew that they were watching a film about the Old West when they saw a gun on the screen, and they were able to distinguish between the film world and the real world. But most of them, as they watched Hart, were probably unaware that they were also seeing the creation of a new art form. The Western film has been the most popular genre of American cinema since the early days of the silent era, and it has had a profound influence on the development of cinema around the world.

22. Knowledge and Progress

Why does the idea of progress loom so large in the modern world? Surely because progress of a particular kind is actually taking place around us and is becoming more and more manifest. Although mankind has undergone no general improvement in intelligence or morality, it has made extraordinary progress in the accumulation of knowledge. Knowledge began to increase as soon as the thoughts of one individual could be communicated to another by means of speech. With the invention of writing, a great advance was made, for knowledge could then be not only communicated but also stored. Libraries made education possible, and education in its turn added to libraries: the growth of knowledge followed a kind of compound - interest law, which was greatly enhanced by

the invention of printing. All this was comparatively slow until, with the coming of science, the tempo was suddenly raised. Then knowledge began to be accumulated according to a systematic plan. The trickle became a stream; the stream has now become a torrent. Moreover, as soon as new knowledge is acquired, it is now turned to practical account. What is called "modern civilization" is not the result of a balanced development of all man's nature, but of accumulated knowledge applied to practical life. The problem now facing humanity is: What is going to be done with all this knowledge? As is so often pointed out, knowledge is a two - edged weapon which can be used equally for good or evil. It is now being used indifferently for both. Could any spectacle, for instance, be more grimly whimsical than that of gunners using science to shatter men's bodies while, close at hand, surgeons use it to restore them? We have to ask ourselves very seriously what will happen if this two - edged weapon falls into the hands of those who would use it for evil purposes. Newspapers and broadcasters, as well as individuals, now spread that information quickly and widely. Disarmament negotiations in Geneva have been going on for years without accomplishing anything. But suppose we succeed in disarming the atom, our work as moral beings is not yet done. It is a problem that calls for very serious consideration. What is the use of disarming the atom if we do not also disarm the mind? It is not enough to take this weapon out of the hands of the soldiers. We must take it out of the hands of the politicians as well. The more knowledge men have, the more responsible they should be. It is only when we are very much more intelligent than we are at present that we shall be able to realize all that knowledge implies.

23. Bird flight

No two sorts of birds practise quite the same sort of flight; the varieties are infinite; but two classes may be roughly seen. Any ship that crosses the Pacific is accompanied for many days by the smaller albatross, Which may keep company with the vessel for an hour without visible or more than occasional movement of wing. The currents of air that the walls of the ship direct upwards, as well as in the line of its course, are enough to give the great bird with its immense wings sufficient sustenance and progress. The albatross is the king of the gliders, the class of fliers which harness the air to their purpose, but must yield to its opposition. In the contrary school, the duck is supreme. It comes nearer to the engines with which man has 'conquered' the air, as he boasts. Duck, and like them the pigeons, are endowed with such-like muscles, that are a good part of the weight of the bird, and these will ply the short wings with such irresistible power that they can bore for long distances through an opposing gale before exhaustion follows. Their humbler followers, such as partridges, have a like power of strong propulsion, but soon tire. You

may pick them up in utter exhaustion, if wind over the sea has driven them to a long journey. The swallow shares the virtues of both schools in highest measure. It tires not, nor does it boast of its power; but belongs to the air, travelling it may be six thousand miles to and from its northern nesting home, feeding its flown young as it flies, and slipping through a medium that seems to help its passage even when the wind is adverse. Such birds do us good, though we no longer take omens from their flight on this side and that; and even the most superstitious villagers no longer take off their hats to the magpie and wish it good - morning.

When Bleriot crossed the Channel at last, it was in a monoplane, an appellation which is now indifferently applied to both the single and double - winged machines, because such machines have by now been so far developed that the initial distinction has no longer any real importance. There is no special branch of standard English for piloting airplanes; still the lingo is so different from that of everyday life that the characters in a play or film have to be made to use some easily recognizable jargon if they are to represent pilots on the stage or screen. So the monoplane became "the thing" with the flying men, a clumsy - looking contrivance of wood and wire and cloth, with something not unlike a box - kite tied in front of it; and so it remained for many years. The word 'aeroplane' was usurped by the newcomers in exactly the same way as the word 'automobile' was usurped by the manufacturers of motor - cars. In England at least, a motor - car was originally a carriage without horses, and the name stuck, even when the horses had been replaced by an engine. The aeroplane, too, at first was an 'air - plane', a structure built to plane in the air, not just to fly as a bird does, or as an insect does, but to glide on a plane, which means, in practice, to float on an air - current, exactly as a sled glides on a snow - current. But the newcomers called their machines 'aeroplanes', and the name has stuck, just as the name 'automobile' has stuck in America. In France, of course, the word 'avion' has always been used, and in Germany the word 'Flugzeug'.

24. Beauty

A young man sees a sunset and, unable to understand or to express the emotion that it rouses in him, concludes that it must be the gateway to a world that lies beyond. It is difficult for any of us in moments of intense aesthetic experience to resist the suggestion that we are catching a glimpse of a light that shines down to us from a different realm of existence, different and, because the experience is intensely moving, in some way higher. And, though the gleams blind and dazzle, yet do they convey a hint of beauty and serenity greater than we have known or imagined. Greater too than we can describe; for language, which was invented to convey the meanings of this world, cannot readily be

fitted to the uses of another.

That all great art has this power of suggesting a world beyond is undeniable. In some moods, Nature shares it. There is no sky in June so blue that it does not point forward to a bluer, no sunset so beautiful that it does not waken the vision of a greater beauty, a vision which passes before it is fully glimpsed, and in passing leaves an indefinable longing and regret. But, if this world is not merely a bad joke, life a vulgar flare amid the cool radiance of the cosmos, and existence an empty laugh braying across the mysteries; if these intimations of a something behind and beyond are not evil humour born of indigestion, or whimsies sent by the devil to mock and madden us, if, in a word, beauty means something, yet we must not seek to interpret the meaning. If we glimpse the unutterable, it is unwise to try to utter it, nor should we seek to invest with significance that which we cannot grasp. Beauty in terms of our human meanings is meaningless.

Beauty is not a need but an ecstasy. It is not a mouth thirsting nor an empty hand stretched forth, but rather a heart enflamed and a soul enchanted. It is not the image you would see nor the song you would hear, but rather an image you see though you close your eyes and a song you hear though you shut your ears. It is not the sap within the leaf, nor the dove upon the bough, but rather the mistral to come, the storm to be weathered. It is not a matter of satisfying simple needs, but rather of fulfilling the highest aspirations of the human spirit.

Beauty is an ecstasy; it is as simple as hunger. There is really nothing to be said about it. It is like the perfume of a flower: you can smell it and that is all. It is a common property of all great art, and one that differentiates great art from merely good art. The more we try to explain it in terms of its why and how, the more it eludes us; yet we know when we have met it, and we know when we have missed it. The best we can do is to describe it, to try to put it into words, but then it is no longer beauty. It is like trying to put your finger on the wind. It is always beyond our grasp, and that is why it is beautiful.

25. Non-auditory Effects of Noise

Many people in industry and the Services, who have practical experience of noise, regard any investigation of this question as a waste of time; they are not prepared even to admit the possibility that noise affects people. On the other hand, those who dislike noise will sometimes use most inadequate evidence to support their pleas for a quieter society. This is a pity, because noise abatement really is a good cause, and it is likely to be discredited if it gets to be associated with bad science. One allegation often made is that noise produces mental illness. A recent article in a weekly newspaper, for instance, was headed with a striking illustration of a lady in a state of considerable distress, with the caption

'She was yet another victim, reduced to a screaming wreck'. On turning eagerly to the text, one learns that the lady was a typist who found the sound of office typewriters worried her more and more until eventually she had to go into a mental hospital. Now the snag in this sort of anecdote is of course that one cannot distinguish cause and effect. Was it the noise, perhaps, that made the woman ill? Or was it the fact that she was already ill that caused her to be hypersensitive to noise? Another patient might equally well complain that her neighbours were combining to slander her and persecute her, and yet one might be cautious about believing this statement. What is needed in the case of noise is a study of large numbers of people living under noisy conditions, to discover whether they are mentally ill more often than other people are. The United States Navy, for some years, has been conducting an experiment along these very lines. They have in view a new weapon that would produce a very loud sound and wondered whether the noise could be used as a weapon of war. So they are now studying the effects of a very loud noise on a large number of people. For this purpose they have set up a colony of human guinea pigs on an island in the Pacific. They have been studying the effects of the noise on these people for several months now. One result of the study has been to show that people do not become mentally ill as a result of noise, but they do become very irritable. There is, in fact, a growing body of evidence to suggest that noise is a contributory factor in a number of diseases, but not necessarily the cause. Another effect of noise is that it reduces efficiency. There are many cases on record of people who, by some extraordinary good luck, have been able to work in the midst of noise without being distracted. But the number of such people is small. Now the reason why noise reduces efficiency is that it distracts people. It makes them think about the noise instead of about their work. For example, a man working in a factory where there is a great deal of noise will be thinking about the noise all the time, and this will make him less efficient. Noise also has a harmful effect on people's sleep. It is a common experience that people who live in noisy environments are likely to have bad sleep. They may not be able to fall asleep at all, or they may be woken up several times during the night. This is because noise disturbs their normal sleep pattern. In conclusion, noise has a number of non - auditory effects on people, and these effects are important and should be taken into account.

26. The past life of the earth

It is animals and plants which lived in or near water whose remains are most likely to be preserved, for one of the necessary conditions of preservation is quick burial, and it is only in the seas and rivers, and sometimes lakes, where mud and silt have been continuously deposited, that bodies and the like can be rapidly covered over and preserved.

But even in the most favourable circumstances only a small fraction of the creatures that die are preserved in this way before decay sets in or, even more likely, before scavengers eat them. After all, all living creatures live by feeding on something else, whether it be plant or animal, dead or alive, and it is only by chance that such a fate is avoided. The remains of plants and animals that lived on land are much more rarely preserved, for there is seldom anything to cover them over. When you think of the innumerable birds that one sees flying about, not to mention the equally numerous small animals like field mice and voles which you do not see, it is very rarely that one comes across a dead body, except, of course, on the roads. They decompose and are quickly destroyed by the weather or eaten by some other creature.

It is almost always due to some very special circumstances that traces of land animals survive, as by falling into inaccessible caves, or into an ice - crevasse, like the Siberian mammoths, when the whole animal is sometimes preserved, as in a refrigerator. This is what happened to the famous Beresovka mammoth which was found preserved and in good condition. In his mouth were the remains of fir - trees - the last meal that he had before he fell into the crevasse some 10,000 years ago. The mammoth has now been restored in the Museum of Zoology in St. Petersburg.

Other animals were trapped in tar pits, like the elephants, sabre - toothed cats, and numerous other creatures that are found at Rancho La Brea, near Los Angeles. Apparently what happened was that water collected on these tar pits, and the animals came to drink. Once they were in the tar, they were stuck fast, and they eventually died there. As the tar pits were gradually filled up by debris, the remains of the animals were buried deeper and deeper, and so were preserved.

There is one fairly common event that sometimes leads to the preservation of land - animal remains, and that is when the animals die in a river or a lake. This may seem a strange statement, but in fact the water in a river or a lake is more likely to be alkaline than acid, and so the bones are more likely to be preserved in such a water. But, even so, the bones of the animals are very likely to be scattered by scavengers and other agents of decay, and only a small fraction of them are likely to be preserved.

One of the most remarkable preservations of land - animal remains has been discovered on the island of Flores in Indonesia. There, on a ledge of rock about 200 metres above sea - level, an almost complete skeleton of a *Homo floresiensis* was found. The skeleton was so small that it was first thought to be that of a child. But when the bones were analysed, it was discovered that they belonged to an adult. The *Homo floresiensis* was a very small human - like creature that lived about 18,000 years ago. It was only about one metre tall, and its brain was only about 400 cubic centimetres, which is about one - third the size of the brain of a modern human. The discovery of the *Homo floresiensis* was a great surprise, for it was thought that all humans had evolved into the modern form long ago.

But the discovery of this small human - like creature shows that evolution can take different paths in different parts of the world.

27. The 'Vasa'

From the seventeenth - century empire of Sweden, the story of a galleon that sank at the start of her maiden voyage in 1628 must be one of the strangest tales of the sea. For nearly three and a half centuries she lay at the bottom of Stockholm harbour until her discovery in 1956. This was the Vasa, royal flagship of the great imperial fleet.

King Gustavus Adolphus, 'The Northern Hurricane', then at the height of his military success in the Thirty Years' War, had dictated her measurements and armament. Triple gun - decks mounted sixty - four bronze cannon. She was intended to play a leading role in the growing might of Sweden.

As she was prepared for her maiden voyage on August 10, 1628, Stockholm was in a ferment. From the Skeppsbron and surrounding islands the people watched this thing of beauty begin to spread her sails and catch the wind. They had laboured for three years to produce this floating work of art; she was more richly carved and ornamented than any previous ship. The high sterncastle was a riot of carved gods, demons, knights, kings, warriors, mermaids, cherubs; and zoomorphic animal shapes ablaze with red and gold and blue, symbols of courage, power, and cruelty, were portrayed to stir the imaginations of the superstitious sailors of the day.

Then the cannons of the anchored warships thundered a salute as the Vasa moved slowly out to sea. She was a moving sight indeed as the morning sun shone on her sails, turning them to gold. As she gained speed, the wind became too strong for her sails, and the mast snapped. Then she lurched sideways and sank in less than a minute.

More than a thousand people lost their lives in the accident; most of them were women and children, who had been on board to enjoy the voyage. The Vasa was the first modern warship built from the keel up with a triple - gun - deck, and she was also the last of her kind. Her sinking was not the result of her being rammed by another vessel, or of her being caught in a storm. It was the result of her own structural weaknesses. The Vasa was built too tall for her length, and she was too top - heavy. Her builders had violated several of the rules of shipbuilding. They had made her too wide for her length, and they had made her sterncastle too high. They had also made her gun - decks too heavy, and her keel too light. All these factors combined to make her unstable in the water. In 1961, the Vasa was salvaged from the harbour bottom, and she is now on display in a museum in Stockholm.

28. Patients and Doctors

This is a sceptical age, but although our faith in many of the things in which our forefathers fervently believed has weakened, our confidence in the curative properties of the bottle of medicine remains the same as theirs. This modern faith in medicines is proved by the fact that the annual drug bill of the Health Services is mounting to astronomical figures and shows no signs at all of ceasing to rise. The majority of the patients attending the medical out-patients departments of our hospitals feel that they have not received adequate treatment unless they are able to carry home with them some tangible remedy in the shape of a bottle of medicine, a box of pills, or a small jar of ointment, and the doctor in charge of the department is only too ready to provide them with these requirements.

There is no quicker method of disposing of patients than by giving them what they are asking for, and since most medical men in the Health Services are overworked and have little time for offering time - consuming and little - appreciated advice on such subjects as diet, right living, and the need for abandoning bad habits etc., the bottle, the box, and the jar are almost always granted them. It is rare for a medical man to give a talk without mentioning the drug companies that finance his research. The more cynical of his profession will even go so far as to claim that "Half of what we are taught about medicine is wrong."

Most doctors, however, still believe that medicine is a science, and they are not prepared to admit that the drugs they prescribe are little more than placebos. They feel that they are being honest when they tell patients that a certain drug may be helpful, but they cannot be certain that it will be. They are also not prepared to admit that the huge sums of money they receive from the drug companies for research are a form of bribery. They believe that their research is for the good of the public and that they are justified in receiving the money.

Few doctors are willing to admit that they are "pill - pushers" and that they prescribe drugs for financial reasons rather than for medical ones. They are afraid that if they do so, they will lose their patients' confidence and that they will no longer be respected as members of a noble profession.

29. The hovercraft

Many strange new means of transport have been developed in our century, the strangest of them being perhaps the hovercraft. In 1953, a former electronics engineer in his fifties, Christopher Cockerell, who had turned to boat - building on the Norfolk Broads, suggested an idea on which he had been working for many years to the British government. It was the idea of supporting a craft on a 'pad', or cushion, of low - pressure air, ringed with a curtain of higher pressure air. Ever since, people have had difficulty in deciding whether the craft should be ranged among ships, planes, or land vehicles - for it is something in between a boat and an aircraft. As a shipbuilder, Cockerell was trying to find a solution to the problem of the wave resistance which wastes a good deal of a surface ship's power and limits its speed. His answer was to lift the vessel out of the water by making it ride on a cushion of air, no more than one or two feet thick. This is done by a great number of ring - shaped air jets on the bottom of the craft. It 'flies', therefore, but it cannot fly higher - its action depends on the surface, water or ground, over which it rides.

The first tests on the Solent in 1959 caused a sensation. The hovercraft travelled first over the water, then mounted the beach, climbed up the dunes, and sat down on a road. Later it crossed the Channel, riding smoothly over the waves, which presented no problem.

Since that time, various types of hovercraft have been developed and some of them have entered regular service on both land and water. The hovercraft is particularly useful in large areas with poor communications such as Africa or Australia; it can become a 'flying fruit - bowl', carrying bananas from the plantations to the ports; giant hovercraft liners could span the Atlantic; and the railway of the future may well be the 'hovertrain', riding on its air cushion over a single rail, which it never touches, at speeds up to 300 m.p.h. - the possibilities appear unlimited.

30. How Animals Migrate

Animal migration, one of the most interesting but least understood biological phenomena, actually refers to the seasonal movement of all or part of an animal population to and from a given area. Familiar migrants include many birds; hoofed animals, especially in East Africa and in the Arctic tundra; bats; whales and porpoises; seals; and fishes, such as salmon.

Migration in some cases is obligatory, and in others facultative. Close and careful study of migratory animals indicates that migratory behavior varies in its adaptability to environmental changes. This variability in migratory behavior has led some scientists to question whether the concept of migration is valid. Certainly the description applied to

some animals is difficult to reconcile with the description applied to others. For example, migratory behavior has been described as "traveling some distance in search of a new habitat" and as "a regular, annual movement between two habitats".

An example of facultative migration is the arctic fox. On the Aleutian Islands, which are located at the western extremity of the Alaskan Peninsula, the fox is sedentary and does not migrate, since there is a relatively stable and abundant food supply on those islands. But in some other parts of the Arctic, the food supply is seasonally limited, and arctic foxes may travel long distances in search of food. This type of movement between different habitats for the purpose of exploiting food resources may be termed facultative migration.

Another example of facultative migration is the large North American deer, or wapiti. In the Rocky Mountains, wapiti migrate seasonally from high - altitude summer ranges to the lower valleys in winter. However, when food is plentiful in the highlands, wapiti may not migrate at all or may migrate only short distances to nearby valleys. Thus, wapiti behavior varies in its adaptability to environmental changes.

The term "obligatory migration" implies that an animal must migrate. One example of obligatory migration is the Atlantic salmon. These fish are born in freshwater streams, and they spend the first few years of their lives in those streams. After that, they migrate to the ocean, where they live for several years. When they reach sexual maturity, they return to their natal streams to spawn. If they are prevented from migrating, they will not reproduce. Another example of obligatory migration is the monarch butterfly. These butterflies are born in North America, and they migrate south to Mexico for the winter. If they are prevented from migrating, they will die.

The reasons for migration are complex. Some animals migrate to find food, while others migrate to find a suitable place to breed. Still others migrate to avoid harsh weather conditions. In general, migration is a strategy that animals use to increase their chances of survival and reproduction.

31. The sculptor speaks

Appreciation of sculpture depends upon the ability to respond to form in three dimensions. That is perhaps why sculpture has been described as the most difficult of all arts; certainly it is more difficult than the arts which involve appreciation of flat forms, shape in only two dimensions. Many more people are 'form-blind' than colour-blind. The child learning to see, first distinguishes only two - dimensional shape; it cannot judge distances, depths. Later, for its personal safety and practical needs, it has to develop

(partly by means of touch) the ability to judge roughly three - dimensional distances. But having satisfied the requirements of practical necessity, most people go no further. Though they may attain considerable accuracy in the perception of flat form, they do not make the further intellectual and emotional effort needed to comprehend form in its full spatial existence.

This is what the sculptor must do. He must strive continually to think of, and use, form in its full spatial completeness. He gets the solid shape, as it were, inside his head - he thinks of it, whatever its size, as if he were holding it completely enclosed in the hollow of his hand. He mentally visualizes a complex form from all round itself; he knows while he looks at one side what the other side is like; he identifies himself with its centre of gravity, its mass, its weight; he realizes its volume, as the space that the shape displaces in the air.

And the sensitive observer of sculpture must also learn to feel shape simply as shape, not as description or reminiscence. He must, for example, perceive an egg as a simple single solid shape, quite apart from its significance as food, or from the literary idea that it will become a bird. And so with solids such as a shell, a nut, a plum, a pear, a tadpole, a mushroom, a mountain peak, a kidney, a carrot, a tree - trunk, a bird's beak, a cloud, a flower. From these he can go on to appreciate more complex forms of combinations of several forms.

The story of the discovery of Troy by Heinrich Schliemann is a good example of the fact that the recognition of the form of an object may be quite independent of its significance. Schliemann, as a child, had heard the story of the destruction of Troy from his father. He became obsessed with the idea that he would find the site of ancient Troy. Years later, when he had made a fortune as a merchant in Russia, he set out on his quest. Guided by the description in Homer's Iliad, he dug at a site in modern Turkey which he thought was the location of Troy. After many years of fruitless digging, he finally found the ruins of a city which he believed was Troy. But what he had really found was a city which had been built on the site of Troy long after the Trojan War. The significance of the discovery was that it showed that the form of the city could be recognized even when its significance had been completely forgotten.

Another example of the independence of form is to be found in the work of the American sculptor Alexander Calder. Calder was famous for his mobiles, which are sculptures that move in the air. Calder's mobiles are made up of simple geometric shapes such as circles, squares, triangles, and rods. These shapes are joined together by wires or rods so that they can move freely in the air. The significance of Calder's mobiles is not in their description of anything in particular, but in their form. They are a celebration of form in movement, of the way in which simple geometric shapes can be combined to create complex and beautiful forms in motion.

32. Galileo Reborn

In his own lifetime Galileo was the centre of violent controversy; but the scientific dust has long since settled, and today we can see even his famous clash with the Inquisition in something like its proper perspective. But, in contrast, it is only in modern times that Galileo has become a problem child for historians of science.

The old view of Galileo was delightfully uncomplicated. He was, above all, a man who experimented: who despised the prejudices and book learning of the Aristotelians, who put his questions to nature instead of to the ancients, and who drew his conclusions fearlessly. He had been the first to turn a telescope to the sky, and he had seen there evidence enough to overthrow Aristotle and Ptolemy together. He was the man who climbed the Leaning Tower of Pisa and dropped various weights from the top, who rolled balls down inclined planes, and then generalized the results of his many experiments into the famous law of free fall.

But a closer study of the evidence, supported by a deeper sense of the period, and particularly by a new consciousness of the philosophical underpinning of seventeenth - century science, has profoundly modified this view of Galileo. Today, although the old Galileo lives on in many popular writings, among historians of science a new and more sophisticated picture has emerged. At the same time, our sympathy for Galileo's opponents has grown. His telescopic observations are justly immortal; they aroused great interest at the time, they had important theoretical consequences, and they provided a striking demonstration of the potentialities hidden in instruments. But can we blame those who looked and failed to see what Galileo saw, if we remember that to use a telescope at the end of the sixteenth century was like playing a modern violin for the first time? Was the philosopher who refused to look through Galileo's telescope more culpable than those who alleged that the spiral nebulae observed by Lord Rosse's great telescope in the eighteen - forties were scratches left by the grinder? We can perhaps forgive those who said that the moons of Jupiter were produced by Galileo's spy - glass if we recall that in his day, as for long afterwards, curved glass was the popular contrivance for producing not truth but illusion, untruth; and if we recall that the concept of an external world was not so much a real concept as a psychological necessity in the minds of seventeenth - century philosophers.

The case of Galileo is sometimes misused to illustrate the thesis that science progresses through the conflict of ideas. But this misrepresents the nature of scientific discovery. The real hero of the story is the slow accretion of the body of knowledge which we call science, and of which Galileo became one of the founders. The most important single aspect of Galileo's work may have been his demonstration of how fundamental questions about the nature of the world could be answered by the use of instruments. Whether his own methods were completely scientific is open to debate; but that he did have a method

is beyond dispute. His method was to make hypotheses, let them stand the test of experiment, and then generalize the results. He was a pioneer in the use of scientific method, which is why we remember him today as one of the great scientists.

33. Education

Education is one of the key words of our time. A man without an education, many of us believe, is an unfortunate victim of adverse circumstances, deprived of one of the greatest twentieth - century opportunities. Convinced of the importance of education, modern states 'invest' in institutions of learning to get back 'interest' in the form of a large group of enlightened young men and women who are potential leaders. Education, with its cycles of instruction so carefully worked out, punctuated by textbooks - those purchasable wells of wisdom - what would civilization be like without its benefits?

So much is certain: that we would have doctors and preachers, lawyers and defendants, marriages and births - but our spiritual outlook would be different. We would lay less stress on 'facts and figures' and more on a good memory, on applied psychology, and on the capacity of a man to get along with his fellow - citizens. If our educational system were fashioned after its bookless past we would have the most democratic form of 'college' imaginable. Among tribal people all knowledge inherited by tradition is shared by all; it is taught to every member of the tribe so that in this respect everybody is equally equipped for life.

It is the ideal condition of the 'equal start' which only our most progressive forms of modern education try to regain. In primitive cultures the obligation to seek and to receive the traditional instruction is binding to all. There are no 'illiterates' - if the term can be applied to peoples without a script - while our own compulsory school attendance became law in Germany in 1642, in France in 1806, and in England in 1876, and is still non - existent in a number of 'civilized' nations. This shows how long it was before we deemed it necessary to make sure that all our children could share in the knowledge accumulated by the 'happy few' during the past centuries.

Education in the wilderness is not a matter of monetary means. All are entitled to an equal start. There is none of the hurry which, in our society, often hampers the full development of a growing personality. There is no fixed programme, no discrimination between one pupil and another in the matter of talents and abilities. At the same as it is true that all pupils are equal in the wilderness, it is also true that all pupils are different. No two persons are the same in terms of abilities, talents, and interests. But in our educational system, all pupils are measured by a single standard - that of scholastic ability. This is the fallacy of the 'equal start' in our schools. The fact that all pupils are different means that

some will learn more quickly than others, some will need more practice than others, some will be more interested in certain subjects than others. But in our schools, all pupils are given the same curriculum, the same textbooks, the same methods of instruction, and the same tests. This is the fallacy of the 'equal start' in our schools.

We may say that, on the whole, the 'equal start' in our schools is a fallacy. But we must also recognize that the fallacy has a certain amount of truth in it. In our modern society, scholastic ability is an important factor in determining a person's success in life. A person with a high scholastic ability is more likely to get a good job, to earn a high salary, to have a high social status, and to be respected by others. But we must also recognize that scholastic ability is not the only factor in determining a person's success in life. There are other factors such as personal qualities, social skills, and work experience. In our modern society, a person with a high scholastic ability but without good personal qualities, social skills, or work experience is not likely to be successful in life. We must also recognize that in our modern society, education is not only a matter of scholastic ability. Education is also a matter of personal qualities, social skills, and work experience. In our modern society, a person who has received a good education is not only a person with a high scholastic ability but also a person with good personal qualities, social skills, and work experience.

34. Adolescence

Parents are often upset when their children praise the homes of their friends and regard it as a slur on their own cooking, or cleaning, or furniture, and often are foolish enough to let the adolescents see that they are annoyed. They may even accuse them of disloyalty, or make some spiteful remark about the friends' parents. Such a loss of dignity and descent into childish behaviour on the part of the adults deeply shocks the adolescents, and makes them resolve that in future they will not talk to their parents about the places or people they visit. Before very long the parents will be complaining that the child is so secretive and never tells them anything, but they seldom realize that they have brought this on themselves.

Disillusionment with the parents, however good and adequate they may be both as parents and as individuals, is to some degree inevitable. Most children have such a high ideal of their parents, unless the parents themselves have been unsatisfactory, that it can, hardly hope to stand up to a realistic evaluation. Parents would be greatly surprised and deeply touched if they realized how much belief their children usually have in their character and infallibility, and how much this faith means to a child. If parents were prepared for this adolescent reaction, and realized that it was a sign that the child was

growing up and developing valuable powers of observation and independent judgment, they would not be so hurt, and therefore would not drive the child into opposition by resenting and resisting it.

The adolescent, with his passion for sincerity, always respects a parent who admits that he is wrong, or ignorant, or even that he has been unfair or unjust. What the child cannot forgive is the parents' refusal to admit these charges if the child knows them to be true.

Victorian parents believed that they kept their dignity by retreating behind an unreasoning authoritarianism. In fact they did nothing of the kind, but children were then too cowed to let them know how they really felt. Today we tend to go to the other extreme, but on the whole this change is a healthy one. However, the authoritarian approach and the laissez - faire approach both have their dangers. The former creates inhibitions and resentments; the latter often results in over - indulgence and lack of discipline. In any case, children have to be prepared for dealing with a world that is full of insecurity, and they have to be given the tools for doing so.

These tools include a healthy self - esteem, which is based on a realistic assessment of one's own abilities; a clear sense of values, which enables a person to make good decisions; and a capacity for empathy, which allows a person to understand the feelings of others. All these things are necessary for a person to be able to function well in the world, and they can be taught to children in a variety of ways. For example, a child can be taught to value his own work by having it displayed or praised; he can be taught to respect the feelings of others by being made to feel what it is like to be in their shoes; and he can be taught to have a realistic view of his own abilities by being given tasks that are appropriate to his age and level of development.

There is no reason why parents should not be able to teach these things to their children. In fact, they are the best people to do so, because they know their children better than anyone else does. However, they must be prepared to be honest with themselves and with their children, and they must be willing to learn as well as to teach.

35. The process of ageing

At the age of twelve years, the human body is at its most vigorous. It has yet to reach its full size and strength, and its owner his or her full intelligence; but at this age the likelihood of death is least. Earlier, we were infants and young children, and consequently more vulnerable; later, we shall undergo a progressive loss of our vigour and resistance which, though imperceptible at first, will finally become so steep that we can live no longer, however well we look after ourselves, and however well society, and our doctors,

look after us. This decline in vigour with the passing of time is called ageing. It is one of the most unpleasant discoveries which we all make that we must decline in this way, that if we escape wars, accidents and diseases we shall eventually 'die of old age', and that this happens at a rate which differs little from person to person, so that there are heavy odds in favour of our dying between the ages of sixty - five and eighty. Some of us will die sooner, a few will live longer - on into a ninth or tenth decade. But the chances are against it, and there is a virtual limit on how long we can hope to remain alive, however lucky and robust we are.

Normal people tend to forget this process unless and until they are reminded of it. We are so familiar with the fact that man ages, that people have for years assumed that the process of losing vigour with time, of becoming more likely to die the older we get, was something self - evident, like the cooling of a hot kettle or the wearing - out of a pair of shoes. They have also assumed that all animals, and probably other organisms such as trees, or even the universe itself, must in the course of time 'wear out'. Most animals we commonly observe do in fact age as we do if they survive long enough; and mechanical systems like a wound - up watch, or the sun, do in fact run out of energy in accordance with the second law of thermodynamics (whether the whole universe does so is a moot point at present). But these are not analogous to what happens when man ages. A run - down watch is still a watch and can be rewound. An old watch, by contrast, becomes so worn and unreliable that it eventually stops working altogether. It is rather like the engine of a car, which can be repaired and even remade, but which eventually is replaced by a new engine. But a human body is not a car, no matter how much you may like to think so. There is no sudden change in the human body which marks the oncoming of old age. The changes are gradual and, at first, are often scarcely perceptible. It is often said that age is a state of mind, and this is true in part. If you feel young, you will be young, but this does not mean that you can avoid the ageing process. The ageing process is a biological process which cannot be avoided, no matter how much you may like to think you can.

The best current medical opinion seems to be that we are born with a sort of 'clock' inside us which determines how long we can live. This 'clock' is thought to be in the genes, and it is believed that we inherit it from our parents. The idea is that the 'clock' starts ticking at birth, and that it keeps on ticking until it runs out of time. Some people believe that this 'clock' can be 'reset' by things like diet, exercise, or drugs, but there is no real evidence for this at present. It is also believed that the rate of ageing can be slowed down by diet, exercise, or drugs, but again, there is no real evidence for this at present. The fact that some people live longer than others may be due to differences in their genes, or it may be due to differences in their lifestyle. It is likely that both factors play a role. But whatever the cause, the fact remains that we all age, and that we all die.

The ageing process has many implications for society. For example, as the population

ages, there will be more and more people who are no longer able to work, and who will need to be supported by society. There will also be more and more people who are ill, and who will need medical care. This will put a strain on the resources of society, both in terms of money and in terms of manpower. Another implication is that as people age, they will become more and more different from each other. Some people will age gracefully, while others will age badly. Some people will be able to cope with the ageing process, while others will not. This will make it more difficult for society to deal with the ageing population as a whole.

In conclusion, the ageing process is a complex and unavoidable biological process which has many implications for society. We need to understand it better if we are going to be able to deal with it effectively.

36. The cost of government

If a nation is essentially disunited, it is left to the government to hold it together. This increases the expense of government, and reduces correspondingly the amount of economic resources that could be used for developing the country. And it should not be forgotten how small those resources are in a poor and backward country. Where the cost of government is high, resources for development are correspondingly low.

This may be illustrated by comparing the position of a nation with that of a private business enterprise. An enterprise has to incur certain costs and expenses in order to stay in business. For our purposes, we are concerned only with one kind of cost - the cost of managing and administering the business. Such administrative overheads in a business are analogous to the cost of government in a nation. The administrative overheads of a business are low to the extent that everyone working in the enterprise realizes clearly what the objectives of the business are and the means of attaining them. If they are not, the enterprise will have to incur extra costs to clarify its objectives. It may have to set up a committee, spend a great deal of time and money on the plan, and then find out that the plan won't work. In the meantime, of course, the business will have been losing money.

If a nation as a whole is to be more efficient, it has to be better organized. Centralized planning, management, and so on, may be highly efficient by themselves, but from the point of view of the nation as a whole we must be careful about charges of inefficiency. There is always another side to the picture. For example, if we look at the matter of centralization of government functions, we find that while centralization may be highly efficient in terms of economic management, it may also lead to a loss of local self-government. We have to keep in mind that in a large nation there will be great differences in the economic development of different regions. If all the economic decisions are made

by a central body, some regions will be left behind. In addition, the large - scale centralization of production that is characteristic of some socialist countries may lead to a loss of the incentives that are provided by a competitive market economy. We have to keep in mind that in a competitive market economy there are always incentives to improve the quality of products and services in order to meet the needs of consumers.

In short, we must be careful when we say that a certain government function, measure, or policy is "inefficient". There may be a complex explanation for the apparent inefficiency. There may be some hidden benefits that we have not yet discovered.

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Normal people tend to forget this process unless and until they are reminded of it. We are so familiar with the fact that man ages, that people have for years assumed that the process of losing vigour with time, of becoming more likely to die the older we get, was something self - evident, like the cooling of a hot kettle or the wearing - out of a pair of shoes. They have also assumed that all animals, and probably other organisms such as trees, or even the universe itself, must in the course of time 'wear out'. Most animals we commonly observe do in fact age as we do if they survive long enough; and mechanical systems like a wound - up watch, or the sun, do in fact run out of energy in accordance with the second law of thermodynamics (whether the whole universe does so is a moot point at present). But these are not analogous to what happens when man ages. A run - down watch is still a watch and can be rewound. An old watch, by contrast, becomes so worn and unreliable that it eventually stops working altogether. It is rather like the human

body. A person who has been doing strenuous physical work for a long time may find that his or her body no longer responds as well as it used to; but if he or she takes a long rest, the body will recover some of its vigour. There is a difference between a worn - out watch and a worn - out body. An old watch cannot be made new again; but a human body can be restored to a certain extent, and, if we take good care of it, it can remain in fairly good condition for a long time.

Our so - called 'control of nature' is a phrase full of conceit. It implies that nature is something outside us, which we can control. We forget that we ourselves are part of nature, and that as we do to nature, nature does to us. Our very life depends on our body's adjustment to the changing environment. We are not much different from other animals in this respect. When we are cold, we shiver and try to get closer to a heat source. When we are hot, we sweat and try to find a cooler place. We do these things because our body has been adjusted to the environment in which we live. We also have a built - in thermostat which regulates our body temperature. When our body temperature rises, we sweat more; when it drops, we shiver more. This is how our body keeps itself in balance. If we interfere with this natural adjustment, for example, by wearing too many clothes when we are cold or by not wearing enough when we are hot, we will get sick. So, we can say that our body has a certain amount of self - control. But this self - control is limited. If we are exposed to too much cold or heat, our body will not be able to adjust itself, and we will die. So, we should respect nature and let our body do its job. We should not be too conceited about our 'control of nature'.

38. Water and the Traveler

Contamination of water supplies is usually due to poor sanitation close to water sources, sewage disposal into the sources themselves, leakage of sewage into distribution systems or contamination with industrial or farm waste. Even if a piped water supply is safe at its source, it is not always safe by the time it reaches the tap. Intermittent tap - water supplies should be regarded as particularly suspect.

Travelers on short trips to areas with water supplies of uncertain quality should avoid drinking tap - water, or untreated water from any other source. It is best to keep to hot drinks, bottled or canned drinks of well - known brand names - international standards of water treatment are usually followed at bottling plants. Carbonated drinks are acidic, and slightly safer. Make sure that all bottles are opened in your presence, and that their rims are clean and dry.

Boiling is always a good way of treating water. Some hotels supply boiled water on request and this can be used for drinking, or for brushing teeth. Portable boiling elements

that can boil small quantities of water are useful when the right voltage of electricity is available. Refuse politely any cold drink from an unknown source.

Ice is only as safe as the water from which it is made, and should not be put in drinks unless it is known to be safe.

Drinking water can be obtained from almost every inhabited building, but the quality should be regarded as suspect, unless it has been treated in some way. If you are worried, buy a water filter from a reliable source and use it. Generally, however, if water has been boiled or treated at a bottling plant, the risk of water - borne disease will be small.

39. Waves

Waves are the children of the struggle between ocean and atmosphere, the ongoing signatures of infinity. Rays from the sun excite and energize the atmosphere of the earth, awakening it to flow, to movement, to rhythm, to life. The wind then speaks the message of the sun to the sea and the sea transmits it on through waves - an ancient, elemental, powerful yet gentle message.

Of all nature's forces, gravity is the most familiar and the most mysterious. Gravity is the irresistible force of attraction that moves objects towards each other. In the case of the ocean and the atmosphere, gravity is the force that keeps the ocean's surface in its place, while allowing the atmosphere to move freely above it. Waves are the result of the interaction between the ocean and the atmosphere, as the wind blows across the surface of the water, creating ripples that grow into waves.

Waves can be described in terms of their height, length, and period. The height of a wave is the vertical distance between the crest (the highest point) and the trough (the lowest point). The length of a wave is the horizontal distance between two consecutive crests or troughs. The period of a wave is the time it takes for one complete cycle of the wave to occur, from crest to crest or trough to trough.

Waves can also be classified according to their source. Wind - generated waves are the most common type of waves, created by the wind blowing across the surface of the water. Tsunamis, on the other hand, are large - scale waves generated by underwater earthquakes, volcanic eruptions, or landslides. Seiches are standing waves that occur in enclosed or semi - enclosed bodies of water, such as lakes or bays.

The power of waves is truly astonishing. Waves can erode coastlines, build up beaches, and transport sediment across great distances. They can also generate electricity, as the movement of the waves can be harnessed to turn turbines. In addition, waves play an important role in the global climate system, as they help to distribute heat around the

world.

In conclusion, waves are a fascinating and important part of nature. They are the result of the interaction between the ocean and the atmosphere, and they possess a wide range of characteristics and functions. Whether we are admiring their beauty from the shore, surfing their crests, or harnessing their power for electricity, waves are an ever - present reminder of the power and mystery of nature.

40. Waves

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Although an ocean wave gives the impression of a wall of water moving in your direction, in actuality waves move through the water leaving the water about where it was. If the water was moving with the wave, the ocean and everything on it would be racing in to the shore with obviously catastrophic results.

An ocean wave passing through deep water causes a particle on the surface to move in a roughly circular orbit, drawing the particle first towards the advancing wave, then up into the wave, then forward with it and then -- as the wave leaves the particles behind -- back to its starting point again.

From both maturity to death, a wave is subject to the same laws as any other 'living' thing. For a time it assumes a miraculous individuality that, in the end, is reabsorbed into the great ocean of life.

The undulating waves of the open sea are generated by three natural causes: wind, earth movements or tremors, and the gravitational pull of the moon and sun. Once waves have been generated, gravity is the force that drives them in a continual attempt to restore the ocean surface to a flat plain.

41. Training elephants

Two main techniques have been used for training elephants, which we may call respectively the tough and the gentle. The former method simply consists of setting an elephant to work and beating him until he does what is expected of him. Apart from any moral considerations this is a stupid method of training, for it produces a resentful animal who at a later stage may well turn man - killer. The gentle method requires more patience in the early stages, but produces a cheerful, good - tempered elephant who will give many years of loyal service.

The first essential in elephant training is to assign to the animal a single mahout who will be entirely responsible for the job. Elephants like to have one master just as dogs do, and are capable of a considerable degree of personal affection. There are even stories of half - trained elephant calves who have refused to feed and pined to death when by some unavoidable circumstance they have been deprived of their own trainer.

The system which seems to work best for elephants is to assign to them their own trainer, who will stay with them for life. Another essential factor is the establishment of a very definite place where the elephant will be trained. It should be a fairly large area, surrounded by a fence or railing, and it should be in an area where the elephant is not likely to be distracted by other animals. Elephants need to be trained to be stable in various situations, so they should be exposed to different environments. They should be trained to stand still when they are being groomed, for example, and to kneel when they are being loaded or unloaded. They should also be trained to respond to commands given in different ways, such as by voice, by whistle, or by a tug on a rope.

Elephants are very intelligent animals, and they can be trained to perform a wide variety of tasks. They can be trained to carry heavy loads, for example, or to work in the logging industry. They can also be trained to perform in circuses, where they can show off their strength and agility. However, it is important to note that elephants are wild animals, and they should be treated with respect and care. Training elephants should be a process that is based on positive reinforcement, rather than on punishment or coercion.

42. Record of the Past

We might marvel at the progress made in every field of study, but the methods of testing a person's knowledge and ability remain as primitive as ever they were. It really is extraordinary that after all these years, educationists have still failed to devise anything more efficient and reliable than examinations. For all the pious claim that examinations test what you know, it is common knowledge that they more often do the exact opposite.

They may be a good means of testing memory, or the knack of working rapidly under extreme pressure, but they can tell you nothing about a person's true ability and aptitude.

As anxiety-makers, examinations are second to none. That is because so much depends on them. They are the mark of success or failure in our society. Your whole future may be decided in one fateful day. It doesn't matter that you weren't feeling very well, or that your mother died. Little things like that don't count: the exam goes on. No one can give of his best when he is in mortal terror, or after a sleepless night, yet this is precisely what the examination system expects him to do.

The moment a child begins school, he enters a world of vicious competition where success and failure are clearly defined and measured. Can we wonder at the increasing number of 'drop - outs': young people who are written off as utter failures before they have even embarked on a career? Can we be surprised at the suicide rate among students?

A good education should, among other things, train you to think for yourself. The examination system does anything but that. What has to be learnt is rigidly laid down by a syllabus, so the student is encouraged to memorize. Examinations do not motivate a student to read widely, but to restrict his reading; they do not enable him to seek more and more knowledge, but induce cramming. They lower the standards of teaching, because they deprive the teacher of all freedoms. Teachers themselves are often judged by examination results, and instead of teaching their subjects, they are reduced to training their students in exam techniques which they despise. The most successful candidates are not always the best educated; they are the best trained in the technique of working under duress.

The results on which so much depends are often nothing more than a subjective assessment by some anonymous examiner. Examiners are only human. They get tired and hungry; they make mistakes. Yet they have to mark stacks of hastily scrawled scripts in a limited amount of time. Most of them are overworked and underpaid. They are in the position of a doctor who has to diagnose a patient's illness without the benefit of laboratory tests. What wonder that in such circumstances, they often make mistakes?

There must surely be many simpler and more effective ways of assessing a person's true abilities. Is it cynical to suggest that examinations are merely a profitable business for the institutions that run them? This is what it boils down to in the last analysis. The best comment on the system is this ill - fated word - 'examination'.

43. Are There Strangers in Space?

We must conclude from the work of those who have studied the origin of life, that given a planet only approximately like our own, life is almost certain to start. Of all the planets in our own solar system we are now pretty certain the Earth is the only one on which life can survive. Mars is too dry and poor in oxygen, Venus far too hot, and so on. But other suns, stars as the astronomers call them, are bound to have planets like our own, and as the number of stars in the universe is so vast, this possibility becomes virtual certainty. There are one hundred thousand million stars in our own Milky Way alone, and then there are three thousand million other Milky Ways, or galaxies, in the universe. So the number of stars that we know exist is now estimated at about 300 million million million.

We can make the same kind of argument, and have in recent years. For example, we think that between 80 and 90 per cent of all stars, like our sun, are circled by planets which may be inhabited. That means that our nearest neighbours are perhaps 4. 3 light years away, which is much too far to think of in terms of visiting them. But even if we never visit them, if we could get at their language, we might be able to communicate with them.

Recently, two Americans made a startling discovery; they have found sound waves in the space between the stars. If such sound waves really exist, they must be the remains of some very large and powerful explosion that took place thousands of millions of years ago. Since these sound waves are spreading outwards at the speed of light, they are by now far out in the universe. And it is also possible that other civilizations have picked up these waves, and are trying to make something of them. This also gives us a new way to look at the universe, and makes us wonder if there are other intelligent beings out there.

44. Patterns of Culture

Custom has not commonly been regarded as a subject of any great moment. The inner workings of our own brains we feel to be uniquely worthy of investigation, but custom, we have a way of thinking, is behaviour at its most commonplace. As a matter of fact, it is the other way around. Traditional custom, taken the world over, is a mass of detailed behaviour more astonishing than what any one person can ever evolve in individual actions, no matter how aberrant. Yet that is a rather trivial aspect of the matter. The fact of first - rate importance is the predominant role that custom plays in experience and in belief, and the very great varieties it may manifest.

No man ever looks at the world with pristine eyes. He sees it edited by a definite set of customs and institutions and ways of thinking. Even in his philosophical probings he cannot go behind these stereotypes; his very concepts of the true and the false will still

have reference to his particular traditional customs. John Dewey has said in all seriousness that the part played by custom in shaping the behaviour of the individual, as against any way in which he can affect traditional custom, is as the proportion of the total vocabulary of his mother tongue against those words of his own baby talk that are taken up into the vernacular of his family. When one seriously studies the social orders that have had the opportunity to develop autonomously, the figure becomes no more than an exact and matter - of - fact observation. The life - history of the individual is first and foremost an accommodation to the patterns of his culture. From his very birth the customs into which he is born shape his experience and behaviour. By the time he can talk, he is the little creature of his culture, and by the time he is grown and able to take part in its activities, its habits are his habits, its beliefs his beliefs, its impossibilities his impossibilities.

Every culture has its own distinctive set of customs and institutions and ways of thinking. These are the building blocks of the culture, and they determine the pattern of the culture. The culture pattern is the overall design of the culture, the way in which the various building blocks are put together. It is the result of the long - term evolution of the culture, and it is relatively stable. Once a culture pattern is established, it is very difficult to change. It is like a mould into which the members of the culture are poured, and they are shaped by it. The culture pattern is not only a set of rules and regulations, but also a set of values and attitudes. It is the way in which the members of the culture see the world and themselves. It is the way in which they evaluate things and people. It is the way in which they make decisions and take actions.

The study of culture patterns is an important part of anthropology. Anthropologists study the different culture patterns of different peoples, and they try to understand how these patterns are formed, how they function, and how they change. They study the relationships between the culture patterns and the social, economic, political, and religious systems of the peoples. They study the impact of the culture patterns on the individual members of the culture, and they study the impact of the individual members on the culture patterns. They try to understand the dynamics of the culture, the way in which it moves and changes. They try to understand the unity and diversity of the culture, the way in which it is both one and many. The study of culture patterns is also an important part of sociology. Sociologists study the culture patterns of different social groups, and they try to understand how these patterns are formed, how they function, and how they change. They study the relationships between the culture patterns and the social structure, the social relations, and the social processes of the social groups. They study the impact of the culture patterns on the individual members of the social groups, and they study the impact of the individual members on the culture patterns. They try to understand the dynamics of the social group, the way in which it moves and changes. They try to understand the unity and diversity of the social group, the way in which it is

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The study of culture patterns is also an important part of psychology. Psychologists study the culture patterns of different peoples, and they try to understand how these patterns are formed, how they function, and how they change. They study the relationships between the culture patterns and the mental processes, the mental states, and the mental characteristics of the peoples. They study the impact of the culture patterns on the individual members of the peoples, and they study the impact of the individual members on the culture patterns. They try to understand the dynamics of the mental processes, the way in which they move and changes. They try to understand the unity and diversity of the mental processes, the way in which it is both one and many.

The study of culture patterns is also an important part of history. Historians study the culture patterns of different peoples in different historical periods, and they try to understand how these patterns are formed, how they function, and how they change. They study the relationships between the culture patterns and the historical events, the historical processes, and the historical trends of the peoples. They study the impact of the culture patterns on the individual members of the peoples, and they study the impact of the individual members on the culture patterns. They try to understand the dynamics of the historical processes, the way in which they move and changes. They try to understand the unity and diversity of the historical processes, the way in which it is both one and many.

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Every child that is born into his group will share them. And no child born into one on the far side of the globe can ever achieve the thousandth part of it. There is no social problem it is more incumbent upon us to understand than this of the role of custom. Until we are able to do so, we will be in danger of violating it in every direction. Only when we understand can we respect. And only when we respect can we hope to modify, if it becomes necessary to modify. But we can't even start to understand until we rid ourselves of the very common notion that custom is only a quaint relic of bygone days.

It is true that in modern times the automatic nature of traditional custom has been much weakened. We no longer live in a world where custom automatically dictates what we do. But the essential role that custom plays in our lives has not changed. We are still creatures of custom, and custom still shapes our experience and beliefs in countless ways.

Custom can never be used as a ready - made solution to modern social problems. But if we are to understand the nature of these problems, we must first understand the role of custom in our lives.

46. Hobbies

A gifted American psychologist has said, "Worry is a spasm of the emotion; the mind catches hold of something and will not let it go." It is useless to argue with the mind in this condition. The stronger the will, the more futile the task. One can only gently insinuate something else into its convulsive grasp. And if this something else is rightly chosen, if it is really attended by the illumination of another field of interest, gradually, and often quite swiftly, the old undue grip relaxes and the process of recuperation and repair begins.

The cultivation of a hobby and new forms of interest is therefore a policy of the first importance to a public man. But this is not a business that can be undertaken in a day or swiftly improvised by a mere command of the will. The growth of alternative mental

interests is a long process. The seeds must be carefully chosen; they must fall on good ground; they must be sedulously tended, if the vivifying fruits are to be at hand when needed.

To be really happy and really safe, one ought to have at least two or three hobbies, and they must all be real. It is no use starting late in life to say: "I will take an interest in this or that." Such an attempt only aggravates the strain of mental effort. A man may acquire great knowledge of topics unconnected with his daily work, and yet get hardly any benefit or relief. It is no use doing what you like; you have got to like what you do. Broadly speaking, human beings may be divided into three classes: those who are toiled to death, those who are worried to death, and those who are bored to death. It is no use offering the manual labourer, tired out with a hard day's work, the chance of playing a game of football or baseball. It is no use inviting the politician or the professional or business man, who has been working or worrying about serious things for six days, to work or worry about trifling things at the weekend.

As for the unfortunate people who can neither toil nor worry, right in the middle of one of their periodic ongosings into boredom, no earthly use can be made of them by anybody. In most cases, the so - called hobbies of the average person are really more like habits. For example, one man's hobby is reading, another's is fishing, another's is gardening, another's is playing the piano. But if you were to ask these people how much time they spend on their hobbies, you would be surprised to find that they spend hardly any time at all. They have the habit of reading, fishing, gardening or playing the piano, but they do not really have the hobby. In fact, they are so busy with their daily work that they have no time to develop a real hobby.

A real hobby is an absorbing pursuit outside one's daily work. It is something that one can lose oneself in. It is not something that one does just to kill time. A real hobby gives one a sense of accomplishment and satisfaction. It is something that one can be proud of. A real hobby can also be a great source of relaxation and rejuvenation.

So, if you want to be really happy and really safe, start early in life to develop a real hobby. Choose something that you are really interested in. Do not choose something just because it is fashionable or because other people are doing it. Develop your hobby slowly but surely. Do not expect too much too soon. And do not give up too easily. If you do these things, you will find that your hobby will become a great source of happiness and safety in your life.

47. The Great Escape

Economy is one powerful motive for camping, since after the initial outlay upon equipment, or through hiring it, the total expense can be far less than the cost of hotels. But, contrary to a popular assumption, it is far from being the only one, or even the greatest. The man who manoeuvres carelessly into his twenty pounds' worth of space at one of Europe's myriad permanent sites may find himself bumping a Bentley. More likely, Ford Escort will be hub to hub with Renault or Mercedes, but rarely with bicycles made for two.

That the equipment of modern camping becomes yearly more sophisticated is an entertaining paradox for the cynic, a brighter promise for the hopeful traveller who has sworn to get away from it all. It also provides - and some student sociologist might care to base his thesis upon the phenomenon - an escape of another kind. The modern traveller is often a man who dislikes the Splendide and the Bellavista, not because he cannot afford, or shuns their material comforts. but because he is afraid of them. Affluent he may be, but he is by no means sure what to tip the doorman or the chambermaid. Master in his own house, he has little idea of when to say boo to a maitre d'hotel.

From all such fears camping releases him. Granted, a snobbery of camping itself, based upon equipment and techniques, already exists; but it is of a kind that, if he meets it, he can readily understand and deal with. There is no superior 'they' in the shape of managements and hotel hierarchies to darken his holiday days.

To such motives, yet another must be added. The contemporary phenomenon of car ownership, which has been mass-produced and which has invaded every aspect of modern life, has helped bring about a situation in which the once - instinctive identification of the person with the place has been weakened. Instead, we have a far more impersonal relationship with the places through which we pass: we view them through car windows at high speed and we are more concerned with the cars in front of us, the cars behind us, the cars to the side. Our sense of place is not the same as that of the ancient hunters and gatherers who knew every rock and tree in their territory. Nor is it the same as that of the peasant who has a deep and personal relationship with the land on which he has worked for generations. In this new situation, camping offers a way of re - establishing a connection with the land, of getting to know a place more intimately than we could from a car window or a hotel balcony. It is a way of getting back to nature, of experiencing the world in a more direct and less artificial way. To some people, camping has become a spiritual as well as a physical escape.

48. Planning a share portfolio

There is no shortage of tipsters around offering "get-rich-quick" opportunities. But if you are a serious private investor, leave the Las Vegas mentality to those with money to fritter. The serious investor needs a proper "portfolio" -- a well-planned selection of investments, with a definite structure and a clear aim. But exactly how does a newcomer to the stock market go about achieving that?

Well, if you go to five reputable stock brokers and ask them what you should do with your money, you're likely to get five different answers, -- even if you give all the relevant information about your age, family, finances and what you want from your investments. Moral? There is no one "right" way to structure a portfolio. However, there are undoubtedly some wrong ways, and you can be sure that none of our five advisers would have suggested sinking all (or perhaps any) of your money into Periwigs.

So what should you do? We'll assume that you have sorted out the basics -- like mortgages, pensions, insurance and access to sufficient cash reserves. You should then establish your own individual aims. These are partly a matter of personal circumstances, partly a matter of psychology.

For instance, if you are older you have less time to recover from any major losses, and you may well prefer to boost your pension income. So you might be more conservative with your investments. On the other hand, if you are younger and can afford to take the long view, you may decide to invest in more speculative stocks, hoping for that big kill in a few years' time.

If you are closer to the "get - rich - quick" end of the spectrum, you will be more interested in speculative stocks that may double or triple in a short time. If you are at the more conservative end, you may prefer bonds or cash deposits. Once you have decided on your investment aims, you can then begin to look for the best investment vehicles to achieve those aims.

The stock market offers a vast array of different investment vehicles, from blue - chip stocks to penny stocks, from bonds to real - estate investment trusts. You need to understand the characteristics of each type of investment vehicle, and how they fit into your overall portfolio. For example, blue - chip stocks are generally considered to be more stable and less risky than penny stocks. Bonds are generally considered to be more conservative than stocks, but they also offer lower returns. Real - estate investment trusts can provide a good income stream, but they are also more sensitive to changes in the real - estate market.

You also need to consider the costs associated with each type of investment vehicle. Some investment vehicles, such as mutual funds, have relatively high management fees. Others, such as exchange - traded funds, have lower management fees. You need to make

sure that the costs associated with your investments do not eat up too much of your returns.

Finally, you need to keep an eye on your portfolio. You should review your portfolio regularly, at least once a year, to make sure that it still meets your investment aims. If your circumstances change, such as if you get married, have a child or change jobs, you may need to adjust your portfolio accordingly.

In conclusion, planning a share portfolio is not an easy task. It requires a good understanding of the stock market, a clear set of investment aims, and a willingness to keep an eye on your investments. But if you are a serious private investor, it is well worth the effort.